

3.2

V je v.p. spojitých reálných fci s $D_f = \langle 1, 5 \rangle$ a skalární součin je definován:

$$\langle f, g \rangle = \int_1^5 f(x) g(x) dx$$

a) ON bází podprostoru $LO\{1, x, x^2\}$ pomocí Gram-Schmidtovy ortogonalizace

$$LO\{1, x, x^2\}$$

$\underbrace{\quad}_{v_1} \quad \underbrace{\quad}_{v_2} \quad \underbrace{\quad}_{v_3}$

$$u_1 = \frac{v_1}{\|v_1\|} = \frac{1}{\sqrt{\langle 1, 1 \rangle}} = \frac{1}{\sqrt{\int_1^5 1 dx}} = \frac{1}{2}$$

$$w_2 = \langle u_1, v_2 \rangle \cdot u_1 = \left\langle \frac{1}{2}, x \right\rangle \cdot \frac{1}{2} = \left\langle \frac{1}{2}, x \right\rangle \cdot \frac{1}{2} = \frac{1}{2} \int_1^5 \frac{x}{2} \cdot dx = 3$$

$$v_2 - w_2 = x - 3$$

$$\|v_2 - w_2\| = \sqrt{\int_1^5 (x-3)^2 dx} = \frac{4}{\sqrt{3}}$$

$$u_2 = \frac{v_2 - w_2}{\|v_2 - w_2\|} = \frac{x-3}{\frac{4}{\sqrt{3}}} = \frac{\sqrt{3}}{4} \cdot (x-3)$$

$$w_3 = \langle u_1, v_3 \rangle \cdot u_1 + \langle u_2, v_3 \rangle u_2$$

$$= \left(\int_1^5 \frac{1}{2} x^2 dx \right) \cdot \frac{1}{2} + \left(\int_1^5 \frac{\sqrt{3}}{4} (x-3) \cdot x^2 dx \right) \cdot \frac{\sqrt{3}}{4} (x-3)$$

$$= \frac{62}{3} \cdot \frac{1}{2} + 32 \cdot \frac{\sqrt{3}}{4} (x-3)$$

$$= 6(x-3) + \frac{31}{3}$$

$$v_3 - w_3 = x^2 - 6(x-3) - \frac{31}{3}$$

$$\|v_3 - w_3\| = \sqrt{\langle v_3 - w_3, v_3 - w_3 \rangle}$$

$$= \sqrt{\int_1^5 \left(x^2 - 6(x-3) - \frac{31}{3} \right)^2 dx}$$

$$= \frac{16\sqrt{5}}{15}$$

$$u_3 = \frac{v_3 - w_3}{\|v_3 - w_3\|} = \frac{15}{16\sqrt{5}} \cdot \left(x^2 - 6(x-3) - \frac{31}{3} \right)$$

$$\text{b\u00e4se } (u_1, u_2, u_3) = \left(\frac{1}{2}, \frac{\sqrt{3}}{4} (x-3), \frac{15}{16\sqrt{5}} \left(x^2 - 6(x-3) - \frac{31}{3} \right) \right)$$



b) Ortogonalní projekce funkce $\sin x$ na tento podprostor, vyjádřenou jako polynom $ax^2 + bx + c$ pro vhodné koeficienty $a, b, c \in \mathbb{R}$

Og projekce $\sin x$ na prostor $P = \langle \underbrace{1}_{u_1}, \underbrace{x}_{u_2}, \underbrace{x^2}_{u_3} \rangle$
je

$$p(x) = ax^2 + bx + c$$

nam ON báz: (u_1, u_2, u_3)

$$(u_1, u_2, u_3) = \left(\frac{1}{2}, \frac{\sqrt{3}}{4}(x-3), \frac{15}{16\sqrt{5}} \left(x^2 - 6(x-3) - \frac{31}{3} \right) \right)$$

Og projekce $\sin x$ na prostor $P = W$

$$W = \langle u_1, \sin x \rangle \cdot u_1 + \langle u_2, \sin x \rangle u_2 + \langle u_3, \sin x \rangle \cdot u_3$$

$$W = \frac{1}{2} \int_1^5 \frac{1}{2} \cdot \sin x \cdot dx + \frac{\sqrt{3}}{4}(x-3) \int_1^5 \frac{\sqrt{3}}{4}(x-3) \cdot \sin x \, dx +$$

$$+ \frac{15}{16\sqrt{5}} \left(x^2 - 6(x-3) - \frac{31}{3} \right) \int_1^5 \frac{15}{16\sqrt{5}} \left(x^2 - 6(x-3) - \frac{31}{3} \right) \cdot \sin x \, dx$$

$W =$

$$\frac{45}{64} x^2 \sin(5) + \frac{45}{64} x^2 \sin(1) - \frac{15}{128} x^2 \cos(5) +$$

$$\frac{15}{128} x^2 \cos(1) - \frac{309}{32} x \sin(5) - \frac{141}{32} x \sin(1) + \frac{21}{64} x \cos(5) -$$

$$\frac{69}{64} x \cos(1) + \frac{309 \sin(5)}{64} + \frac{381 \sin(1)}{64} - \frac{3 \cos(5)}{128} + \frac{291 \cos(1)}{128}$$

$$\begin{aligned}
 W = & \left(\frac{45}{64} \sin 5 + \frac{45}{64} \sin 1 - \frac{15}{128} \cos 5 + \frac{15}{128} \cos 1 \right) \cdot x^2 \\
 & + \left(-\frac{129}{32} \sin 5 - \frac{141}{32} \sin 1 + \frac{21}{64} \cos 5 - \frac{69}{64} \cos 1 \right) x \\
 & + \left(\frac{309}{64} \sin 5 + \frac{381}{64} \sin 1 - \frac{3}{128} \cos 5 + \frac{291}{28} \cos 1 \right)
 \end{aligned}$$

$$W = -0,05x^2 - 0,33x + 5,99$$

c) Norma rozdílu $\|\sin x - (ax^2 + bx + c)\|$

$$\|\sin x - (ax^2 + bx + c)\| = \sqrt{\langle \sin x - (ax^2 + bx + c), \sin x - (ax^2 + bx + c) \rangle}$$

$$= \sqrt{\int_1^5 (\sin x - (ax^2 + bx + c)) dx}$$

$$\begin{aligned}
 = & \sqrt{\int_1^5 \left(\sin x - \left(\frac{45}{64} \sin 5 + \frac{45}{64} \sin 1 - \frac{15}{128} \cos 5 + \frac{15}{128} \cos 1 \right) \cdot x^2 \right.} \\
 & + \left(-\frac{129}{32} \sin 5 - \frac{141}{32} \sin 1 + \frac{21}{64} \cos 5 - \frac{69}{64} \cos 1 \right) x \\
 & \left. + \left(\frac{309}{64} \sin 5 + \frac{381}{64} \sin 1 - \frac{3}{128} \cos 5 + \frac{291}{28} \cos 1 \right) \right) dx}
 \end{aligned}$$

$$= \sqrt{\int_1^5 (\sin x - (-0,05x^2 - 0,33x + 5,99))^2 dx}$$

$$= 11,01$$